Occupancy Dataset
Data Quality Report

Source: Occupancy.csv from archive (4).zip | Audit date: October 11, 2026

Overall finding

The dataset is structurally complete and passes the basic validity checks performed. No missing cells, exact duplicate rows, duplicate timestamps, or invalid occupancy labels were identified. Two recording gaps and unusual sensor readings require contextual review. These findings do not establish that occupancy labels are incorrect.

Dataset overview	Result
Observations / columns	20,560 rows / 7 columns
Time coverage	February 2, 2015, 14:19:00 to February 18, 2015, 09:19:00
Variables	date, Temperature, Humidity, Light, CO2, HumidityRatio, Occupancy
Label distribution	0: 15,810 (76.90%) | 1: 4,750 (23.10%)
Source handling	Original observations preserved; no deletion, imputation, or relabeling.

1. Structure, completeness and duplicates

Check	Finding
CSV structure	Every record has 7 fields; no duplicate headers.
Missing / blank / nonfinite	0 missing cells; 0 blank strings; 0 nonfinite numeric values.
Data types	Dates parse successfully; five continuous sensors are numeric; Occupancy is integer. No constant columns.
Exact rows / timestamps	0 exact duplicate rows; 0 duplicate timestamps.
Repeated sensor patterns	1,441 excess repeated sensor-and-label combinations when timestamp is excluded.
Conflicting labels	0 rows with identical five-sensor combinations but different occupancy labels.

Repeated sensor readings at different timestamps are not necessarily duplicate observations. Stable conditions and sensor rounding can produce repeated values; removing them automatically could alter the temporal data.

2. Sensor ranges and statistical outliers

Sensor	Minimum	Maximum	IQR flags
Temperature	19.000	24.408	519
Humidity	16.745	39.500	0
Light	0.000000	1,697.250	183
CO2	412.750	2,076.500	1,449
HumidityRatio	0.002674	0.006476	0

Method: Flag continuous sensor readings below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR, where IQR = Q3 - Q1. This is a statistical screening rule, not proof of sensor failure.

2,022 distinct rows (9.83%) have at least one IQR flag. Column counts overlap and must not be added as a count of affected rows. Occupancy was excluded because it is a binary class label; applying IQR filtering to it would incorrectly flag the minority occupied class.

No humidity values outside 0-100%, negative light, nonpositive CO2, or negative HumidityRatio were found. Temperature values appear plausible for indoor measurements if expressed in Celsius. Light has many zero readings, which makes a global outlier rule less informative.

3. Time continuity

Records are chronological. Ordinary timestamp intervals are 59, 60, or 61 seconds. Two gaps exceed 90 seconds:

Gap	Start / end	Duration	Unobserved slots*
1	Feb 4, 2015 10:43 Feb 4, 2015 17:51	7h 8m	427
2	Feb 10, 2015 09:33 Feb 11, 2015 14:48	29h 15m	1,754

*Approximate absent one-minute observations assuming continuous collection. The total is 2,181 slots. These are not missing cells in existing rows; they may represent intentional breaks between recording sessions. Do not interpolate across these gaps without confirming the collection schedule.

4. Occupancy consistency review

Exploratory rule	Rows
Occupancy = 1 and Light = 0	1
Occupancy = 0 and Light > 400	185
Occupancy = 0 and CO2 > 1,000	1,148

All Occupancy values are 0 or 1, with no missing or fractional labels. The interpretation 0 = unoccupied and 1 = occupied is assumed. The 400 and 1,000 thresholds are exploratory choices, not documented occupancy rules. An empty room can have daylight or lights left on, and elevated CO2 may persist after departure. These rows should be reviewed, not automatically relabeled.

5. Implications for machine learning

Finding	Possible effect / practical response
Unusual sensor-label pairs	Valid examples can help the model learn realistic exceptions. Incorrect labels could reduce performance. Confirm label provenance before changing them.
Class imbalance	Predicting unoccupied for every row gives 76.90% accuracy across this file. Evaluate occupied-class precision, recall and F1, balanced accuracy, and a confusion matrix.
Adjacent repeated readings	Random row splitting can place very similar minutes in training and test sets, potentially inflating scores. Prefer chronological or recording-session splits appropriate to the intended use.
Outliers	Extreme values can influence scale-sensitive models. Inspect sensor limits and nearby readings; fit any scaling, clipping or learned filtering using training data only.
Recording gaps	Avoid creating lag or rolling features across recording breaks. Any time-derived features should use only information available at prediction time.

Measured impact: No ML model was trained in this audit. The effect on predictive performance remains unknown. A follow-up sensitivity comparison could use the same untouched chronological test set to evaluate training on all records versus a justified alternative treatment of reviewed records.

6. Recommended next steps

1. Preserve the source and convert date to datetime for analysis.
2. Confirm sensor units, instrument ranges, collection sessions, and how occupancy labels were recorded.
3. Inspect flagged readings alongside nearby timestamps and available observation logs.
4. Retain plausible extremes and repeated observations; correct or exclude only confirmed errors with documented reasons.
5. Establish a chronological baseline model and evaluate both occupancy classes before deciding whether additional cleaning is helpful.

Scope and limitations

This audit checked all supplied records, CSV field counts, column names and types, missingness, duplicates, time ordering and gaps, basic logical bounds, binary occupancy validity, and continuous-variable IQR extremes. It did not verify source labels against human observations or certify instrument accuracy. Units are inferred, not supplied: Celsius, relative humidity percent, lux, ppm, and humidity ratio in kg/kg. Exact psychrometric consistency was not assessed because pressure and calculation assumptions are unknown.

Across statistical and contextual checks combined, **2,547 distinct rows (12.39%)** were flagged for review. This is a review workload, not an error rate. Detailed records, original CSV line numbers, column profiles, gap details, and reproducible audit code are included in the previously delivered Occupancy_Quality_Audit.zip.